FAKE NEWS DETECTION AI

Hybrid NLP, Transformer and LLM-Based Fake News Classification System

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Course:
Practical Application of AI

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Abstract

This project presents a comprehensive fake news detection system that combines traditional machine learning, transformer-based deep learning, and large language models (LLMs) to identify misinformation in news articles. The system implements three distinct approaches: (1) Logistic Regression with TF-IDF vectorization as a baseline, (2) Fine-tuned DistilBERT transformer model for contextual understanding, and (3) an ensemble system that combines the local DistilBERT model with Groq LLaMA 3.3 70B for enhanced reliability and explainability. A FastAPI-based REST API with an integrated web interface was developed to make the system accessible. The models were trained on a dataset of approximately 44,000 news articles from Kaggle, achieving up to 99.4% accuracy on the test set. The ensemble approach provides not only classification results but also confidence scores and natural language explanations, making it suitable for real-world applications.

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1. Introduction

1.1 What is Fake News?

Fake news refers to fabricated information that mimics the format of legitimate news media but contains deliberately misleading or false content. In the digital age, the rapid spread of misinformation through social media platforms and online news outlets has become a significant societal challenge.

1.2 Importance of Fake News Detection

The proliferation of fake news poses serious threats to democratic processes, public health, and social cohesion. During critical events such as elections or pandemics, misinformation can influence public opinion, undermine trust in institutions, and even endanger lives. Automated detection systems are essential tools in combating this phenomenon.

1.3 Project Objectives

The primary objectives of this project are:

- Develop a multi-model fake news detection system combining traditional ML and deep learning approaches
- Implement a transformer-based model (DistilBERT) for contextual text understanding
- Integrate a Large Language Model (LLaMA 3.3 70B) via Groq API for ensemble decision-making
- Create a user-friendly web interface and REST API for practical usage
- Achieve high accuracy while providing explainable predictions

2. Problem Definition

2.1 Challenges in Misinformation Detection

Detecting fake news presents several unique challenges. Unlike spam detection, fake news often employs sophisticated language that mimics legitimate journalism. The content may contain partial truths, emotional manipulation, and references to real events or people, making binary classification difficult.

2.2 Need for AI-Based Solutions

Manual fact-checking cannot keep pace with the volume of content generated daily. AI-based solutions offer scalable, consistent, and increasingly accurate alternatives. By leveraging natural language processing (NLP) techniques, machine learning models can learn patterns indicative of misinformation from large labeled datasets.

3. Dataset Description

3.1 Dataset Source

The dataset used in this project is the "Fake and Real News Dataset" available on Kaggle (clmentbisaillon/fake-and-real-news-dataset). It contains two CSV files: Fake.csv and True.csv.

3.2 Dataset Statistics

Metric	Value
Total Articles	44,898
Fake Articles	23,481
Real Articles	21,417

After preprocessing (removing duplicates and short texts), 39,103 articles were used for training and evaluation. The dataset was split into training (80%), validation (10%), and test (10%) sets with stratified sampling.

4. Methodology

4.1 TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) is a statistical measure used to evaluate the importance of a word in a document relative to a collection of documents. It converts text data into numerical vectors that machine learning algorithms can process. In this project, TF-IDF was used with scikit-learn's TfidfVectorizer with English stop words removal and a maximum document frequency threshold of 70%.

4.2 Logistic Regression

Logistic Regression serves as the baseline traditional machine learning model. Despite its simplicity, it achieved 97.4% accuracy on the test set when combined with TF-IDF features. The model was implemented using scikit-learn's Pipeline for clean and reproducible code.

4.3 DistilBERT Architecture

DistilBERT is a smaller, faster, and lighter version of BERT (Bidirectional Encoder Representations from Transformers). It retains 97% of BERT's language understanding capabilities while being 60% faster and having 40% fewer parameters. The model was fine-tuned on the news dataset for 2 epochs with a batch size of 16 and a maximum sequence length of 256 tokens.

4.4 Groq LLaMA 3.3 70B Integration

LLaMA 3.3 70B is a large language model accessible via the Groq API. It provides deep contextual understanding and natural language reasoning capabilities. In this project, LLaMA analyzes news articles independently and provides classification results along with explanatory text.

4.5 Ensemble Decision System

The ensemble system combines predictions from both DistilBERT and LLaMA 3.3 70B. When both models agree, the confidence score is boosted. When they disagree, the system reports the disagreement and provides both predictions for human review. This approach increases reliability and provides transparency.

5. System Architecture

The system follows a modular architecture with clear separation of concerns:

- Data Layer: Dataset preparation and preprocessing scripts
- Model Layer: DistilBERT and Logistic Regression implementations
- Service Layer: Predictor service handling ensemble logic
- API Layer: FastAPI endpoints for external access
- Presentation Layer: Web interface for user interaction

The predictor service first attempts to load the DistilBERT model. If unavailable, it falls back to the Logistic Regression model. For each prediction request, both the local model and the LLM (if configured) analyze the text, and their results are merged into a final response.

6. Implementation

6.1 FastAPI Backend

FastAPI was chosen as the web framework due to its high performance, automatic validation, and built-in OpenAPI documentation. The application is structured with separate modules for schemas, services, and models.

6.2 API Endpoints

Endpoint	Method	Description
/	GET	Web interface for testing
/health	GET	System status check
/predict	POST	Analyze news text and return result

6.3 Swagger Documentation

Interactive API documentation is automatically generated by FastAPI and available at /docs. It allows users to test endpoints directly from the browser.

6.4 Web Interface

A responsive web interface was built using vanilla HTML, CSS, and JavaScript. Users can paste news text, click "Analiz Et" (Analyze), and view results including classification, confidence score, and LLM explanation.

7. Results

7.1 Model Comparison

Model	Dataset	Accuracy	F1 Score
Logistic Regression	Kaggle 44k	97.4%	0.974
DistilBERT	Kaggle 44k	99.4%	0.994

7.2 Confidence Score Explanation

The confidence score represents the system's certainty in its prediction. When both models agree, the confidence is high (typically above 90%). When models disagree, the confidence is adjusted downward to reflect uncertainty. The LLM also provides a natural language explanation for its decision, enhancing the system's transparency.

8. Screenshots

Figure 1: Swagger API Documentation

Figure 1 shows the automatically generated Swagger documentation for the Fake News Detection API. It displays all available endpoints (/health, /predict) and their request/response schemas.

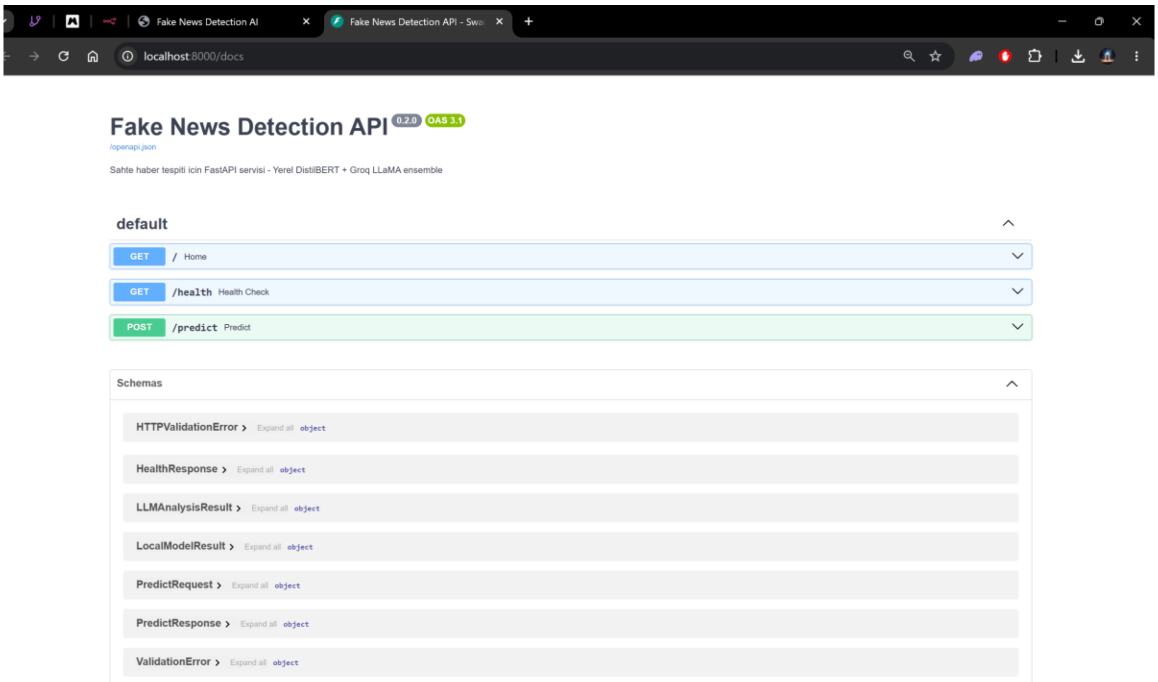


Figure 2: Web Interface - Fake News Detection

Figure 2 demonstrates the web interface detecting a fake news article. The input text contains sensational claims about NASA and aliens. The system correctly classifies it as FAKE with 99% confidence. Both DistilBERT and LLaMA 3.3 70B agree on the classification.

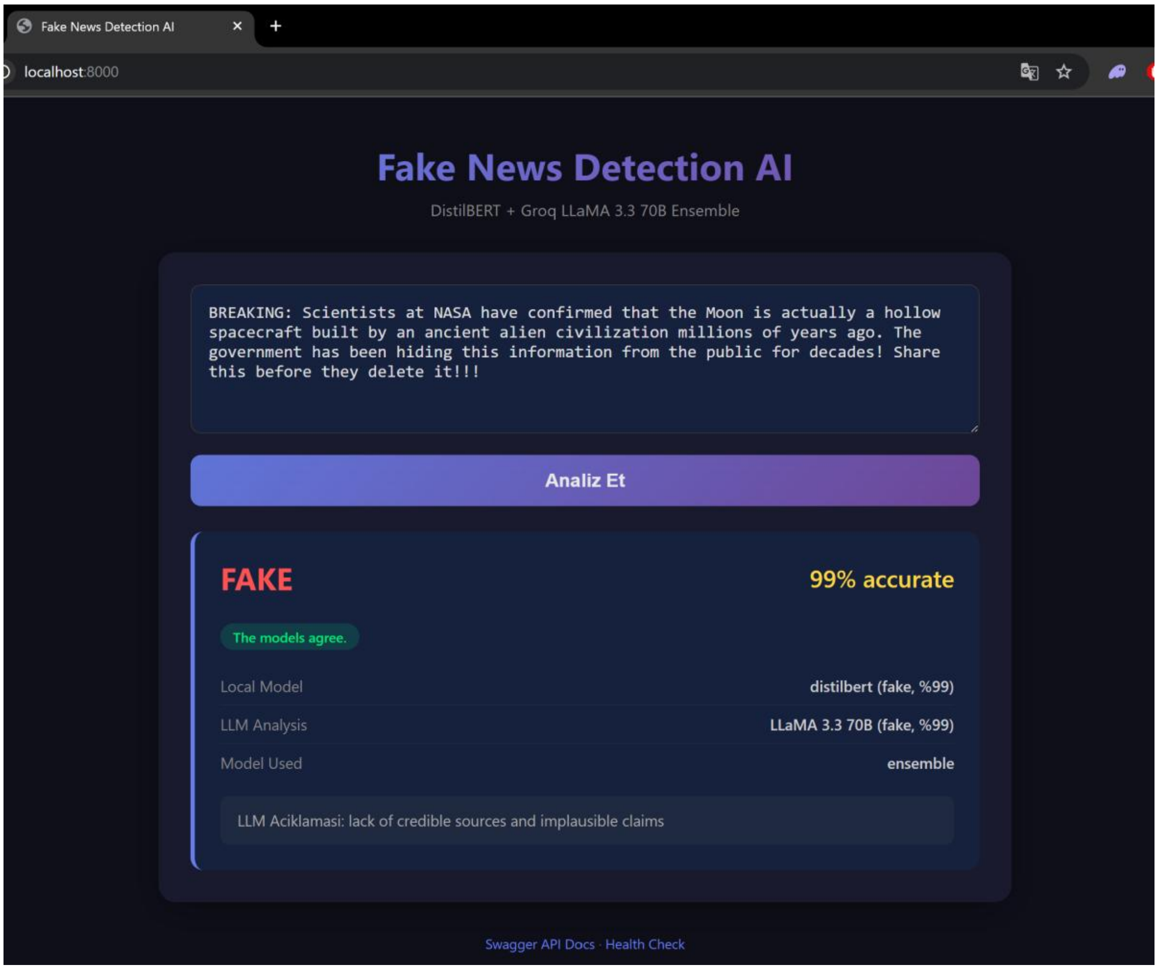


Figure 3: Web Interface - Real News Detection

Figure 3 shows the system correctly classifying a real news article about the European Central Bank. The ensemble system identifies it as REAL with 99% confidence. The LLM provides an explanation citing the credible source and typical financial news reporting style.

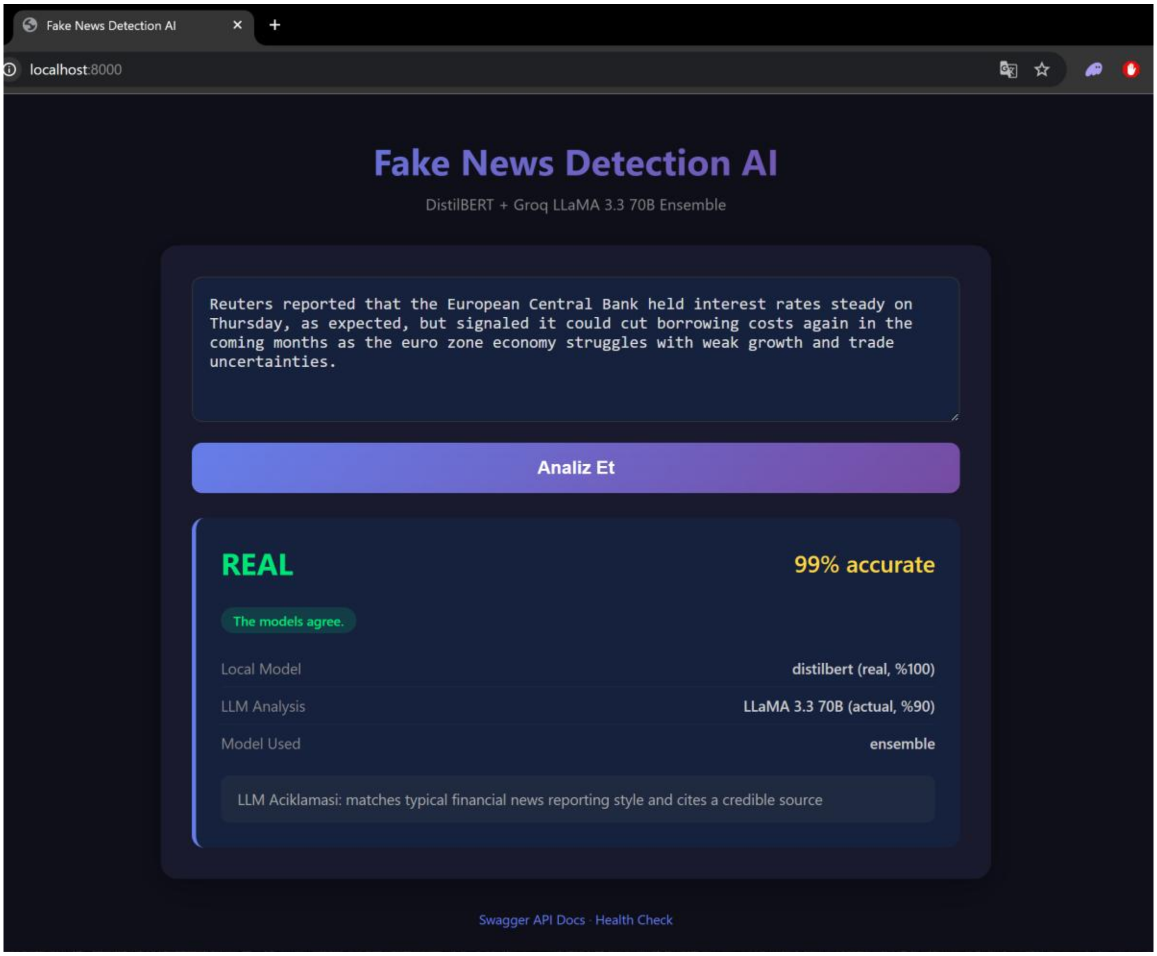
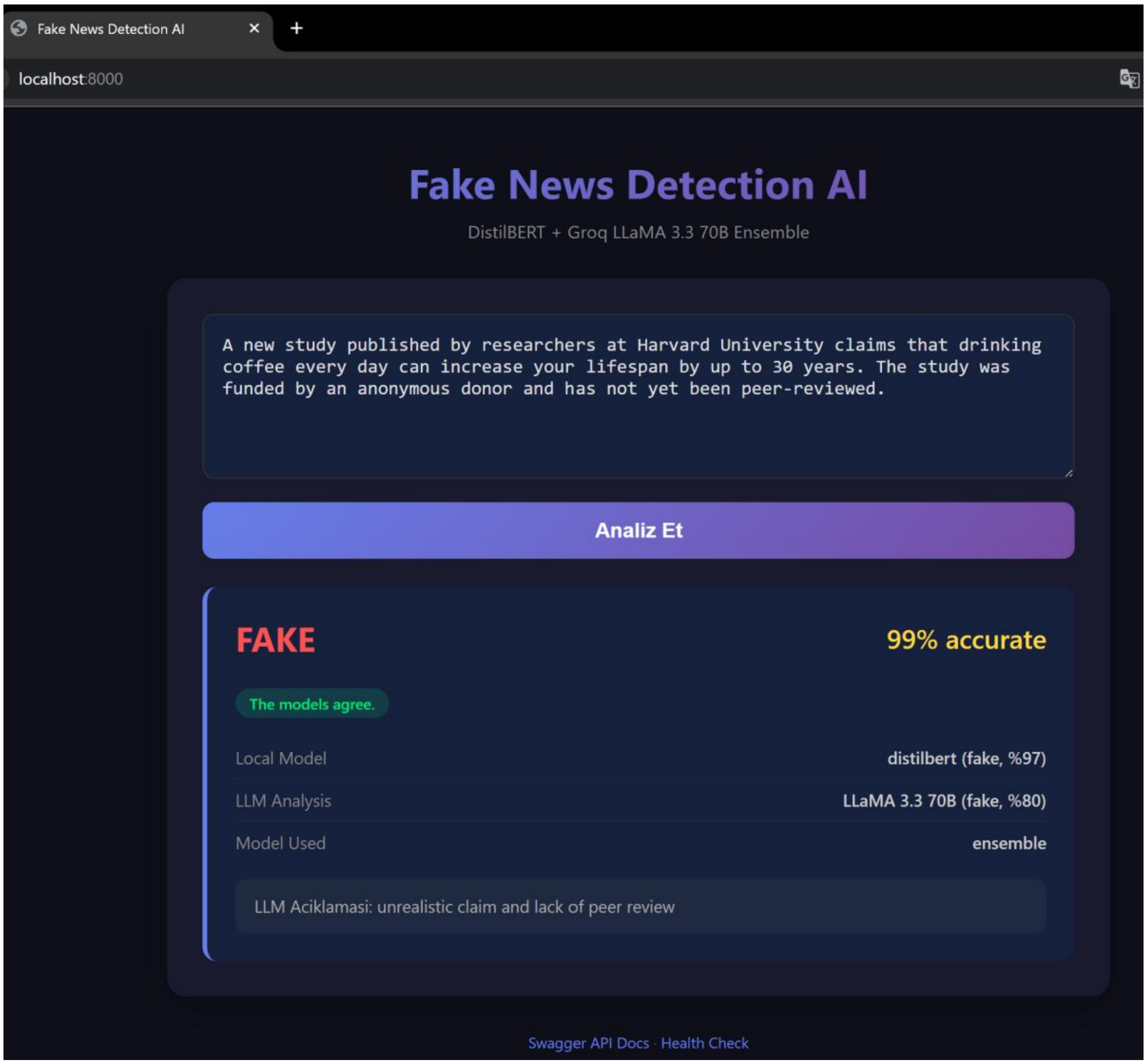


Figure 4: Web Interface - Mixed Content Analysis

Figure 4 presents an analysis of a Harvard University study claim. The system detects it as FAKE with 99% confidence. The LLM explanation highlights the unrealistic claim and lack of peer review, demonstrating the system's ability to provide reasoning for its decisions.



9. Discussion

9.1 Advantages

- Multi-model approach increases reliability and reduces false positives
- Ensemble system provides confidence scores and explanations
- FastAPI backend ensures high performance and scalability
- Web interface makes the system accessible to non-technical users
- Modular architecture allows easy extension and maintenance

9.2 Limitations

- Requires internet connection for LLM ensemble (Groq API)
- Training DistilBERT is computationally expensive on CPU
- Model performance may vary on different types of misinformation
- Currently supports only English text

9.3 Practical Applications

- Social media platforms for content moderation
- News aggregators for filtering unreliable sources
- Educational tools for media literacy training
- Government agencies for monitoring misinformation campaigns

10. Conclusion

This project successfully demonstrates the effectiveness of combining traditional machine learning, transformer models, and large language models for fake news detection. The DistilBERT model achieved 99.4% accuracy on the test set, while the ensemble system with LLaMA 3.3 70B provides enhanced reliability and explainability. The FastAPI backend and web interface make the system practical for real-world deployment.

The importance of AI in combating misinformation cannot be overstated. As digital content continues to grow exponentially, automated detection systems like this one will play a crucial role in maintaining information integrity and supporting informed decision-making.

Author Declaration

This project was independently designed, implemented, trained, tested, and documented by Muhammet Mustafa Buğra Aslan (Student ID: 20264).

All code, models, and documentation were created as part of the Practical Application of AI course at Sarajevo University (UNSA), Department of DSAI.

11. Future Work

- Multilingual Support: Extend the system to detect fake news in multiple languages
- Cloud Deployment: Deploy the API to cloud platforms (AWS, Azure, GCP) for wider accessibility
- Real-time Fact Checking: Integrate with live news feeds for real-time misinformation detection
- Social Media Analysis: Extend the system to analyze social media posts and comments
- Continuous Learning: Implement online learning to adapt to new misinformation patterns
- Mobile Application: Develop a mobile app for on-the-go fake news detection

12. References

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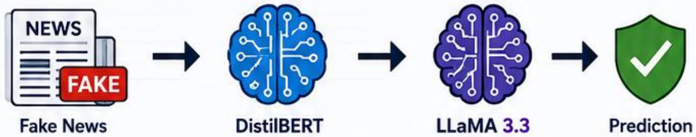


- Accurate Detection
- AI-Powered Analysis
- Explainable Results
- Real-time Prediction

ABSTRACT

The rapid spread of misinformation on the internet has increased the need for automated fake news detection systems. This project presents a hybrid artificial intelligence solution that combines traditional Machine Learning, Transformer-based Deep Learning, and Large Language Models (LLMs) to classify news articles as Real or Fake.

The system integrates Logistic Regression, DistilBERT, and Groq LLaMA 3.3 70B into a unified architecture and provides predictions through a FastAPI web application.

OBJECTIVES

- Detect fake news automatically
- Compare traditional ML and Transformer models
- Integrate LLM reasoning capabilities
- Develop a scalable REST API
- Provide explainable predictions

DATASET

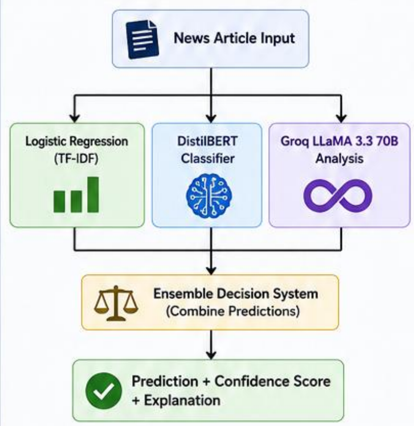
Source: Fake and Real News Dataset (Kaggle)

Dataset	Articles
Fake News	~23,000
Real News	~21,000
Total	~44,000

Preprocessing Steps:

- Text Cleaning
- Tokenization
- Lowercasing
- TF-IDF Vectorization
- Train/Test Split

SYSTEM ARCHITECTURE



MODEL PERFORMANCE

Advantages of Ensemble System

- Improved reliability
- Better contextual understanding
- Reduced prediction uncertainty
- Explainable AI outputs

TECHNOLOGIES

- Python
- FastAPI
- Hugging Face Transformers
- PyTorch
- Scikit-Learn
- Groq API
- Pandas
- NumPy

IMPLEMENTATION

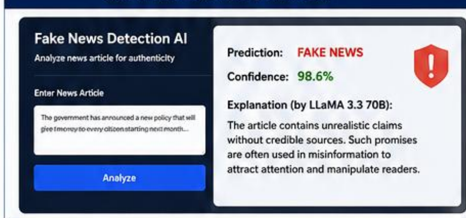
REST API Endpoints

- POST /predict
- GET /health

- FastAPI Backend
- Interactive Swagger Documentation
- Web-based User Interface
- Real-time Prediction
- AI-generated Explanations
- Modular Project Structure

FastAPI — Swagger Supported

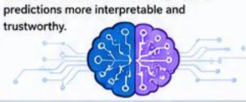
USER INTERFACE PREVIEW



RESULTS

The proposed system successfully detects fake news with high accuracy by combining Machine Learning and Transformer-based approaches.

The integration of Groq LLaMA provides additional contextual reasoning, making predictions more interpretable and trustworthy.



FUTURE WORK

- Multi-language support
- Real-time news verification
- Social media misinformation detection
- Advanced ensemble optimization
- Deployment to cloud platforms

GITHUB REPOSITORY
github.com/Mmbugraaslan/fake-news-detection-ai

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SCAN ME
Check out the project on GitHub!